

FUSIONIX '26 · #TRACK-02 ECOENERGY SYSTEMS

EcoShare

Surplus to Sustain

A zero-waste cascade that connects campus food surplus to people, animals — and energy. Smart bins **sense waste in real time**, and the platform turns every kg into meals, animal feed, and biomethane fuel.

67M

TONNES WASTED YEARLY

194M

UNDERNOURISHED INDIANS

60M

STRAY ANIMALS STARVING

Press → or click to explore

⚠ THE CRISIS

India wastes **67 million tonnes** of food every year

Enough to feed every malnourished person in the country three times over.

Meanwhile, food rotting in landfills produces **methane — 25x more potent than CO₂** — a wasted energy source that no one is capturing.

₹92,000 Cr

VALUE OF FOOD WASTED ANNUALLY

8-10%

GLOBAL EMISSIONS FROM FOOD WASTE

25x

METHANE VS CO₂ POTENCY

💡 The problem isn't just hunger — it's wasted energy. And nobody's solving both together.

THE SOLUTION

Introducing the **Zero-Waste Cascade**

Every kg of food finds its highest purpose — people, animals, or fuel. Nothing reaches landfill.



Feed People

Edible surplus food is matched with volunteers, orphanages, and communities in under 30 minutes.



Feed Animals

Stale rotis, veg peels, overripe fruits — routed to shelters, Gaushalas, and street feeders.



Biogas Fuel

Truly rotten waste goes to anaerobic digesters → biomethane → runs vehicles & generators.



Compost

Nothing wasted — the final residue becomes nutrient-rich compost for campus gardens.

COMPONENT ARCHITECTURE

Four layers. One closed loop.

From waste bin to biogas pump — every kg is sensed, routed, and tracked end-to-end.

● Layer 1 • Sensing (bin-side)

Ultrasonic HC-SR04 reads fill level, HX711 load cell weighs the bin, DHT11 tracks temperature & humidity for food safety, MQ-4 detects methane leaks.

ESP32 NodeMCU

HC-SR04

HX711

DHT11

MQ-4

Solar + Li-ion



● Layer 2 • Connectivity

ESP32's onboard Wi-Fi streams readings over HTTPS to Firebase — with offline queuing and deep-sleep for low power.

Wi-Fi

HTTPS

Firebase

Deep-sleep



● Layer 3 • Cascade Intelligence

Rule engine decides every kg's fate — edible → people, animal-safe → shelters, rotten → biogas. Temperature & expiry logic auto-triggers pickups.

Rule engine

Temp logic

Demand match

Alerts



● Layer 4 • EcoShare Platform

Live dashboard, real-time charts, campus map, notifications, and a judge panel — all from a single-file web app.

Dashboard

Charts

Judge panel

● ⚡ Energy Closed-Loop

Non-edible waste → anaerobic digester → biomethane → campus vehicles & generators.

Biogas plant

Biomethane

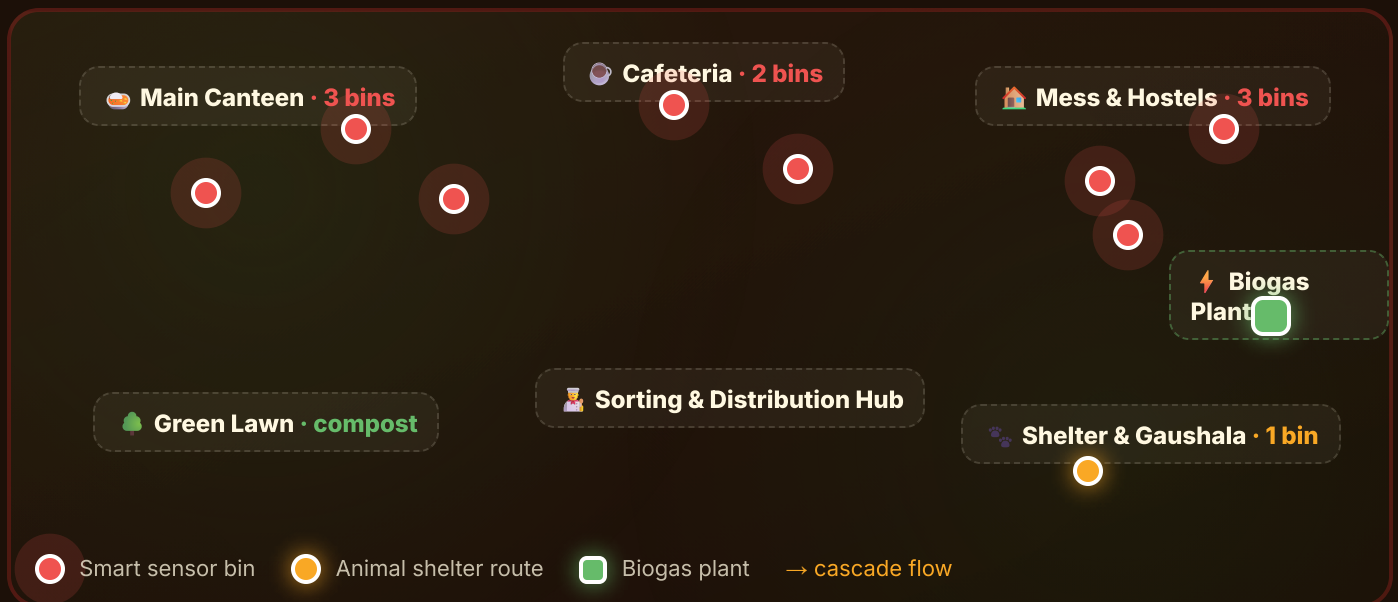
0 grid emissions

1 kg waste → 0.6 m³ biogas → 1.5 km car travel

PROJECTED SENSOR MAP

A sensing net across UEM Kolkata

Projected deployment of 12 smart bins — each streaming fill level, weight, temperature & gas data live to the EcoShare cloud.



🎯 SUSTAINABILITY TARGETS

Targets we can prove

Concrete, measurable goals for the 24-hour build and 8-week campus pilot.

90%

WASTE DIVERTED FROM LANDFILL



Campus food waste rerouted to people, animals & energy instead of the dump.

3,000

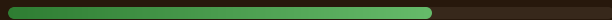
MEALS RESCUED / SEMESTER



Edible surplus matched to shelters & communities through the live feed.

800 m³

BIOGAS → 2,000 KM FUEL



Anaerobic digestion converts rotten waste into vehicle-grade biomethane.

4,500 kg

CO₂-EQ AVOIDED



Capturing methane (25× CO₂) that would otherwise leak from landfill.

₹150/km

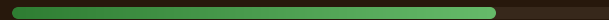
FUEL COST SAVED VS PETROL



Biomethane displaces fossil fuel in campus transport & generators.

12

LIVE SENSOR NODES · 99% UPTIME



ESP32 smart bins streaming real-time data with solar + battery backup.

24h build

2-node working prototype

4-week pilot

12 smart bins · 1 campus

Scale

Multi-campus + city biogas

 **LIVE PLATFORM**

Built for **real-time** impact

Hardware on the ground. Software in the cloud. Everything wired together.



Smart Bin Telemetry

ESP32 nodes stream fill level, weight & temperature to the dashboard live



Smart Cascade

Auto-routes food to the right tier — human, animal, or biogas



Dashboard

Chart.js visualizations: pie chart, bar chart, interactive campus map



Gamification

Leaderboard with badges — Green Warrior, Animal Hero, Energy Pioneer



Judge Panel

Admin dashboard with activity log & exportable impact reports



Safety Monitor

Temperature tracking & expiry alerts for perishable food batches

WHY WE'RE DIFFERENT

No one else does **all three**

Existing apps only feed humans. EcoShare covers the full cascade — and powers it with hardware.

Feature	Existing Apps	EcoShare
Coverage	Only feed humans	Humans + Animals + Biogas
Approach	Ad-hoc donations	Smart cascade matching
Hardware	None	ESP32 sensor mesh → energy
Tracking	No impact tracking	Charts, maps, leaderboard
Focus	General purpose	Built for campus ecosystem

📈 IMPACT METRICS

Every kg counts

Real-time tracking of the difference we're making together.



0+

MEALS SAVED FROM TRASH



0 kg

CO₂ OFFSET



0

ANIMALS FED



0 L

BIOGAS GENERATED

100 kg waste → 150 km car 🚗

BIOGAS CONVERSION RATE

</> TECH STACK

Hardware to cloud. **Zero build.** Instant impact.

ESP32 sensing nodes feed a single-file web platform — deployed to Firebase.

ESP32 NodeMCU

Smart-bin controller, built-in Wi-Fi

HC-SR04 · HX711

Fill level + weight sensing

DHT11 · MQ-4

Temp, humidity, methane

Firebase

Auth, Firestore, realtime sync

HTML/CSS + Tailwind

Indian folk single-file UI

Vanilla JS

All logic, state, Canvas 2D

Chart.js + GSAP

Charts & transitions

Solar + Li-ion

Off-grid, zero-emission nodes

ROADMAP

From prototype to **campus-scale net-zero**

Proving the cascade on one campus, then scaling it across the city.



Fleet Scaling

12 → 200 solar-powered smart bins



Solar Nodes

Zero grid draw, fully off-grid sensing



LoRaWAN

Long-range telemetry for wide campuses



Predictive Fill

ML-driven pickup & demand forecasting



City Biogas

Tie-in with municipal biogas plants



Govt Programs

PMGKAY, PM Poshan, SAMPADA



Thank You

Let's build a zero-waste future.

"When we waste food, we waste the land, water, energy —
and the chance to feed someone."

 **Live Demo: eco-share-26.web.app**

Team EcoShare | FusioniX '26 | #TRACK-02 ECOENERGY SYSTEMS